

IDENTIFYING CHALLENGES IN DESIGNING, DEVELOPING AND EVALUATING DATA VISUALIZATIONS FOR LARGE DISPLAYS

Mahsa Sinaei Hamed*, Pak Kwan, Matthew Klich, Jillian Aurisano, Fateme Rajabiyazdi

*Corresponding Author: mahsasinaeihamed@cmail.carleton.ca

INTRODUCTION

As data sizes grow, visualizing them becomes more complex. Large displays (LDs) offer the necessary space for showcasing such visualizations. However, designing and developing for LDs present distinct challenges.

OBJECTIVE

We aim to gain insights into the challenges designers and developers faced while creating data visualizations for LDs.



METHODOLOGY

Conducted a series of semi-structured interviews with 4 experts in creating data visualizations for LDs. Through affinity diagramming, categorized the challenges they encountered when creating data visualizations for LDs.

RESULTS

Findings identified preliminary challenges in 4 categories, including limited resources for building infrastructure, scaling data visualization designs, limited development resources, and evaluating data visualizations with people.

Themes	Limited Resources for Building LD Infrastructure	Scaling Data Visualization Designs for LDs	Limited Development Resources and Tools for LDs	Evaluating Data Visualizations on LDs with People
Subthemes	Allocating Sufficient Space Allocating proper Cooling Systems Difficulty Relocating LDs	LDs Differ in Size LDs Differ in Aspect Ratio	Developing away from LDs Specialized Interaction Techniques Limited Tools and Software Prototyping	People being unfamiliar with LD Keyboard-to-Screen Workflow Evaluation Studies in the Wild with LDs
Representative Quotes	"We moved all the chairs and tables ... making people to collaborate and use the LD to explore the data." (P1)	"What we did, ... we had a wall and like 4 flipchart ... to actually simulate [the LD]." (P3)	"I didn't get too much help from ... just in that because this is like a very specific platform that I needed." (P4)	"We grew up with this thing that you don't want to break the TV, ... people are afraid to [touch] them." (P3)

NEXT STEPS

Conducting more interviews with experts in creating data visualizations for LDs and extending our understanding of the challenges encountered by data visualization experts while creating data visualizations for LDs.

ACKNOWLEDGMENTS

This work was funded by the Canadian NSERC fund RGPIN-2021-04222.